

P3-7 TEST EVIDENCE

Image

Test ID	Test	Expected Result	Evidence
IT - P1	Encode and decode the short Learning Outcome message using a normal PNG, 2-bit LSB and correct settings.	Authentic, Signature Valid, Media hash Yes, Message hash Yes, start pixel recovered.	PAYLOAD SIZE 7984 bits AVAILABLE CAPACITY 540000 bits STATUS Fits Embed & Sign Public key copied to clipboard. Cover - click to choose start  Exact LSB Change Map  Stego  Stego Heat Map  EMBED RESULT LSB depth 2 Capacity 540000 bits Embedded 7984 bits Changed pixels 1330 Start pixel (68, 145) Authentic DECODED MESSAGE hello RESULTS Signature Valid Payload protection AES-256-GCM Media hash match Yes Embedded message hash 2cf24dba5fb0a30e26e83b2ac5b9e29e1b161e5c1fa7425e73043362938b9824 Decoded message hash 2cf24dba5fb0a30e26e83b2ac5b9e29e1b161e5c1fa7425e73043362938b9824 Message hash match Yes Start location Recovered pixel (68, 145) Details OK

IT - P2

Encode and decode the large Project Overview paragraph using a normal PNG and 4-bit LSB.

LAB-P3-7

1 Image Encoder

Non-verified data inside a PNG image

COVER IMAGE

Cover PNG: 

Choose...

Message / note

Media ID

Message SHA-256 (before embedding)

LSB depth (1-8)

Start location secret

Selected start pixel

☒ Encrypt custom payload (AES-256-GCM)

Public key (share with verifier)

CAPACITY CHECK

LAB-P3-7

2 Image Decoder

Recover the payload from a PNG image

COVER IMAGE

Stego PNG: 

Choose...

Media ID

LSB depth (1-8)

Start location secret

Signer public key (PKM)

Verify: Authentic

Recovered message

RESULTS

Signature

Payload protection

Media hash match

Embedded message hash

Decoded message hash

Message hash match

Start location

Details

IT - N1

Attempt to embed the large Project Overview paragraph into the 32 x 32 PNG at 1-bit LSB.

Too Large; embedding rejected and no output saved.

Message / note

Load from .txt

Message SHA-256 (before embedding)

LSB depth (1-8)

Start-location secret

Selected start pixel

☒ Encrypt custom payload (AES-256-GCM)

Generate Signing Keypair

Public key (share with verifier)

Copy

CAPACITY CHECK

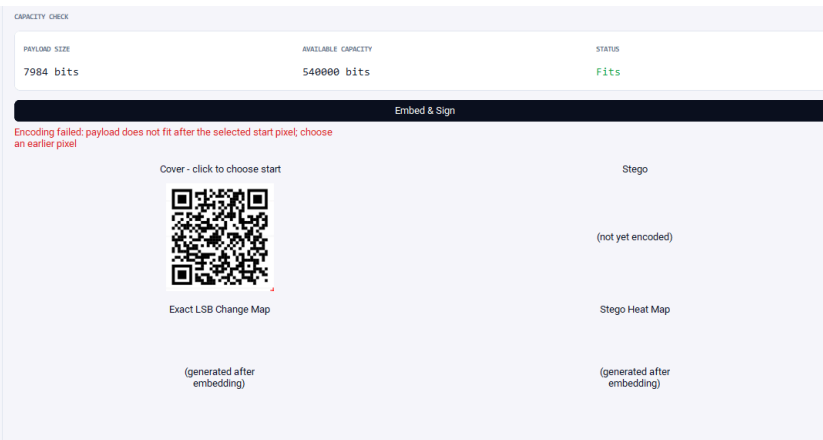
Embed & Sign

Keypair generated for this session (private key kept in memory only).

Cover - click to choose start

Stego

(not yet encoded)

IT - N2	Verify the protected PNG using an incorrect start-location secret	Payload cannot be located; GUI shows Payload Not Found/Payload Missing.	
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Audio

Test ID	Test	Expected Result	Evidence
AT - P1	WAV + short payload + 1 LSB	Encode succeeds; verify Authentic	PAYLOAD & SETTINGS Media ID audio-20261001T131608Z Message / note 7xKSmQ9IzP2vW#4bN9&L1mX5qZ7vR9\$Y3xB8vN2mK5qL1vX8zP3wB9\$KT2mN5qL8vX4zP1wB7\$K13mN9qX8vR2zP4wB7\$K1mN5qL8vX2zP9wB3\$K17mN2qX8vR4zP9wB1\$K1M5mN2qL8vX7zP3wB9\$K11mN5qX8vR2zP4wB7\$K1M9mN1qL8vX2zP3wB7\$K19mN5qX8vR2zP4wB7\$K1mN5qL8vX2zP9wB3\$K17mN2vY8vB4vD0wB1\$K1L5mN7n18vY7vP2wB0\$K11mN5vY8vB7vD4wB7\$K1M0mN1n18vY7vP2wB7\$K1 LSB depth (1-8) 1 2 3 4 5 6 7 8 Start-location secret **** ☒ Encrypt custom payload (AES-256-GCM) Generate Signing Keypair Public key (share with verifier) Copy -----BEGIN PUBLIC KEY----- MCowBQYDK2VwAyEAJF+Fku5ZwtebUGcW8FeaT3ytiqhplwHfVg1GR03rvm= -----END PUBLIC KEY----- CAPACITY CHECK PAYLOAD SIZE 3884 bytes AVAILABLE CAPACITY 166069 bytes STATUS Fits Embed & Sign Encoded encrypted and signed payload successfully: test_audio_embedded.wav – 15580 samples changed. WAV STEGO DIAGNOSTICS

AT - P2	WAV + larger payload + 4 LSB	Encode succeeds; verify Authentic	Media ID audio-202610011131608Z Message / note 9mN5qX8vR2zP4wB7SkM1mN5qL8vX2zP9wB3SkL7mN2qX8vR4zP9wB1SkM5mN2qL8vX7zP3wB9SkL1mN5qX8vR2zP4wB7SkM9mN1qL8vX2zP3wB7SkL9mN5qX8vR2zP4wB7SkM1mN5qL8vX2zP9wB3SkL7mN2qX8vR4zP9wB1SkM5mN2qL8vX7zP3wB9SkL1mN5qX8vR2zP4wB7SkM9mN1qL8vX2zP3wB7SkL0mN5qY8vR7zP4wB7SkM1mN5qL8vY7zP0wB2z3mM0nY8vR7zP4wB7SkM1mN5qL8vY7zP0wB2z3SkL7m LSB depth (1-8) 1 2 3 4 5 6 7 8 Start-location secret **** ☒ Encrypt custom payload (AES-256-GCM) Generate Signing Keypair Public key (share with verifier) Copy -----BEGIN PUBLIC KEY----- MCowBQYDK2VwAeApsISzDdAKY1S1j1Dt8eYd1G08b6i/1n8jw2i/2x/Ts= -----END PUBLIC KEY----- CAPACITY CHECK PAYLOAD SIZE 102468 bytes AVAILABLE CAPACITY 664277 bytes STATUS Fits Embed & Sign Encoded encrypted and signed payload successfully: test_audio_embedded.wav - 192140 samples changed.
AT - P3	FLAC cover → FLAC stego	Encode succeeds; verify Authentic	Cover audio flac_test.flac Browse... COVER INFO CHANNELS 2 SAMPLE RATE 44100 Hz DURATION 233.2 s Play Cover Play Stego Stop PAYLOAD & SETTINGS Media ID audio-202610011131608Z Message / note 9mN5qX8vR2zP4wB7SkM1mN5qL8vX2zP9wB3SkL7mN2qX8vR4zP9wB1SkM5mN2qL8vX7zP3wB9SkL1mN5qX8vR2zP4wB7SkM9mN1qL8vX2zP3wB7SkL9mN5qX8vR2zP4wB7SkM1mN5qL8vX2zP9wB3SkL7mN2qX8vR4zP9wB1SkM5mN2qL8vX7zP3wB9SkL1mN5qX8vR2zP4wB7SkM9mN1qL8vX2zP3wB7SkL0mN5qY8vR7zP4wB7SkM1mN5qL8vY7zP0wB2z3mM0nY8vR7zP4wB7SkM1mN5qL8vY7zP0wB2z3SkL7m LSB depth (1-8) 1 2 3 4 5 6 7 8 Start-location secret **** Cover audio flac_test.flac Browse... Stego audio flac_embedded.flac Browse... Verification status Play Stego Stop Verification status Authentic

Play Cover

Play Steps

Stop

PAYLOAD & SETTINGS

Media ID

audio-20261001I132507Z

Message / note

7xk\$mq9lzf2vW#4bn9&kL1mX5qz7vr9\$ty3xB8vN2mK5qL1vx8zP3wB9\$kt2mNsQl8vX4zP1wB7\$skL3mn9qX8vRzP4wB7\$SkM1mNsQl8vXz2P9wB3\$skL7mN2qX8vR4zP9wB1\$SkM5mN2qL8vXz2P3wB9\$skL1mNsQl8vRzP4wB7\$SkM9mN1qlX5qz2P3wB7\$skL9mNsQl8vRzP4wB7\$SkM1mNsQl8vXz2P9wB3\$skL7mN2qX8vR4zP9wB3\$skL4f5mNmTnl8vY7P3uBdD\$11mM5vYB-dP7vB4uB7\$J4MdmM1nl8vY7P3uB7\$J4

LSB depth (1-8)

12345678

Start-location secret

☒

Encrypt custom payload (AES-256-GCM)

Generate Signing Keypair

Public key (share with verifier)

Copy

-----BEGIN PUBLIC KEY-----
McovBYDK2VwaYEa0hAFLF4tzh2nttKCvGvAX3oXu62TBvOsteInJyGv2c=
-----END PUBLIC KEY-----

CAPACITY CHECK

PAYLOAD SIZE

AVAILABLE CAPACITY

STATUS

11172 bytes

11025 bytes

Too large

Embed & Sign

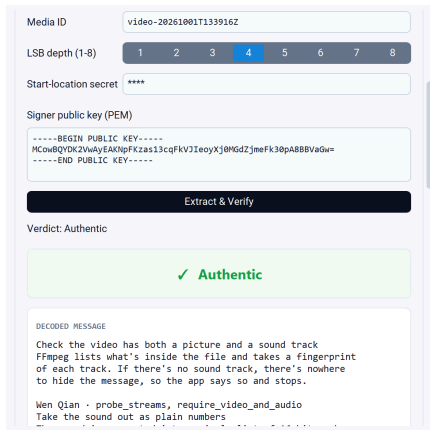
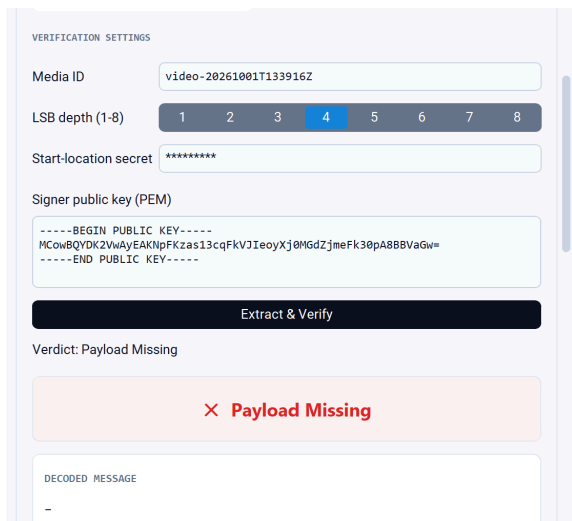
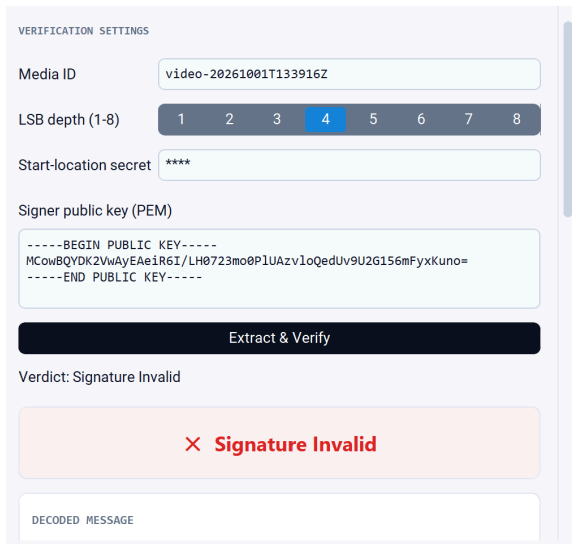
Payload does not fit: needs 11172 bytes, audio has 11025 bytes at 1-bit LSB depth.

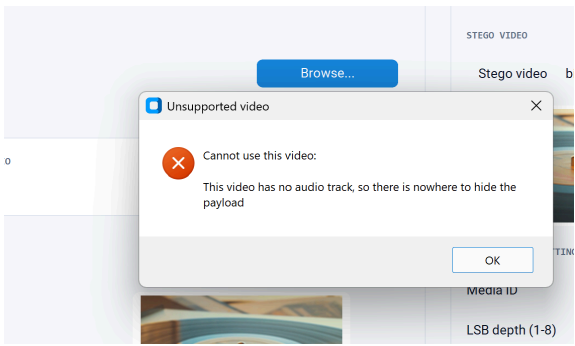
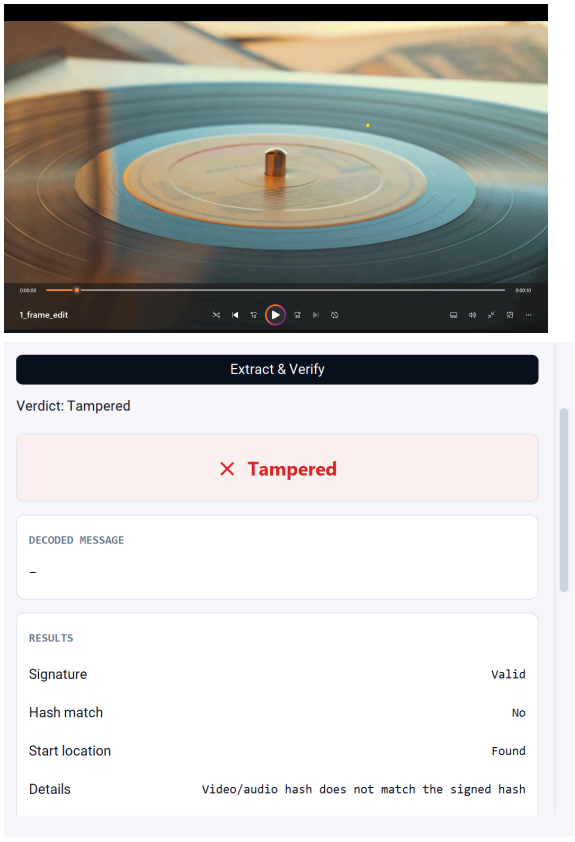
ATN - 3	Wrong start secret	Payload Missing	Stego audio: test1.wav Play Stego Stop VERIFICATION SETTINGS Media ID: test1 LSB depth (1-8): 1 Start-location secret: ***** Signer public key (PEM):-----BEGIN PUBLIC KEY-----MCowBQYDK2VwAyEAeG3CqVc3eo4Juo40tNrtjpPwFFKtIwrkrXmVw61ePBw-----END PUBLIC KEY----- Extract & Verify Verdict: Payload Missing ✗ Payload Missing DECODED MESSAGE - RESULTS Signature: - Payload protection: - Hash match: - Start location: - Details: locator header magic mismatch - no payload found Activate Windows Go to Settings to activate Windows.
ATN - 4	Plain cover with no payload	Payload Missing	Stego audio: test_audio.wav Play Stego Stop VERIFICATION SETTINGS Media ID: test1 LSB depth (1-8): 1 Start-location secret: **** Signer public key (PEM):-----BEGIN PUBLIC KEY-----MCowBQYDK2VwAyEAeG3CqVc3eo4Juo40tNrtjpPwFFKtIwrkrXmVw61ePBw-----END PUBLIC KEY----- Extract & Verify Verdict: Payload Missing ✗ Payload Missing DECODED MESSAGE - RESULTS Signature: - Payload protection: - Hash match: - Start location: - Details: locator header magic mismatch - no payload found Activate Windows

Video Test Cases (Optional Challenge): Toh Wen Qian

ID: VT = Video Test · P = Positive (should pass > Authentic) · N = Negative (should be caught or rejected)

Test ID	Case	Expected Result	Evidence						
VT - P1	Embed a short message into a 10-second MP4 at 5-bit LSB, then verify with the correct media ID, secret and public key.	Authentic; the self-check passes, the picture is unchanged, and the hidden message is displayed.	Encode LSB depth (1-8) 12345678 Start-location secret **** Generate Signing Keypair Public key (share with verifier) Copy -----BEGIN PUBLIC KEY----- MCowBQYDK2VwYyEAd3ovmIhAksqRveQDwFAGtzay93ckoP8V3kaVdKJPqHq= -----END PUBLIC KEY----- CAPACITY CHECK <table><tr><td>PAYLOAD SIZE</td><td>AVAILABLE CAPACITY</td><td>STATUS</td></tr><tr><td>640 bytes</td><td>611840 bytes</td><td>Fits</td></tr></table> Embed & Sign Encoded and self-checked: video_sample_decode.mp4 – 989 audio samples changed. Video stream untouched. Decoded  Open Stego in Player VERIFICATION SETTINGS Media ID video-20261001T133916Z LSB depth (1-8) 12345678 Start-location secret **** Signer public key (PEM) -----BEGIN PUBLIC KEY----- MCowBQYDK2VwYyEAd3ovmIhAksqRveQDwFAGtzay93ckoP8V3kaVdKJPqHq= -----END PUBLIC KEY----- Extract & Verify Verdict: Authentic ✓ Authentic	PAYLOAD SIZE	AVAILABLE CAPACITY	STATUS	640 bytes	611840 bytes	Fits
PAYLOAD SIZE	AVAILABLE CAPACITY	STATUS							
640 bytes	611840 bytes	Fits							
VT - P2	Embed the large Project Overview paragraph into the same MP4 at 4-bit LSB and verify it.	Authentic; the self-check passes, the picture is unchanged, and the hidden message is displayed.	Message / note Check the video has both a picture and a sound track Ffingp lists what's inside the file and takes a firm account of audio tracks. If there's no sound track then Load from .txt LSB depth (1-8) 12345678 Start-location secret **** Generate Signing Keypair Public key (share with verifier) Copy -----BEGIN PUBLIC KEY----- MCowBQYDK2VwYyEAd3ovmIhAksqRveQDwFAGtzay93ckoP8V3kaVdKJPqHq= -----END PUBLIC KEY----- CAPACITY CHECK <table><tr><td>PAYLOAD SIZE</td><td>AVAILABLE CAPACITY</td><td>STATUS</td></tr><tr><td>4276 bytes</td><td>489472 bytes</td><td>Fits</td></tr></table> Embed & Sign Encoded and self-checked: bpg_payload.mp4 – 8021 audio samples Decoded:	PAYLOAD SIZE	AVAILABLE CAPACITY	STATUS	4276 bytes	489472 bytes	Fits
PAYLOAD SIZE	AVAILABLE CAPACITY	STATUS							
4276 bytes	489472 bytes	Fits							

			
VT - N1	Verify the stego video using a wrong start-location secret.	Payload Missing; the secret points to the wrong position, so no header is found.	
VT - N2	Verify the stego video using a valid public key from a different keypair.	Signature Invalid; the signature was not made by the matching private key.	

VT - N3	Attempt to use an MP4 with no audio track as the cover.	Rejected; a pop-up explains there is no audio track to hide the payload in, and nothing is embedded.	
VT - N4	Subtle frame edit 1 small yellow square for 1 second, audio unchanged	Tampered (the picture's SHA-256 changes, so the fingerprint no longer matches.)	

VT - N5	Audio Edit (0.1s replaced)	Tampered (the payload is found, but the audio fingerprint no longer matches.)	Verdict: Tampered ✕ Tampered DECODED MESSAGE - RESULTS Signature Valid Hash match No Start location Found Details Video/audio hash does not match the signed hash
VT - N6	Extra audio track added	Tampered (the file no longer has exactly one video and one audio track.)	Verdict: Tampered ✕ Tampered DECODED MESSAGE - RESULTS Signature - Hash match No Start location Not found Details :ontain exactly one video stream and one audio stream (found: ['v', 'a', 'a'])
VT - N7	Trim + Re-export	Payload Missing (lossy compression destroys the hidden bits.)	Verdict: Payload Missing ✕ Payload Missing DECODED MESSAGE - RESULTS Signature - Hash match - Start location Not found Details locator header magic mismatch – no payload found